COMPLEXITY CLASSES

- In Computer Science, many problems are solved where the objective is to maximize or minimize some values, whereas in other problems we try to find whether there is a solution or not.
- The problems can be categorized into

I. Optimization Problem
II. Decision Problem

Optimization Problem

- Optimization problems are those for which the objective is to maximize or minimize some values.

Example

- Finding the minimum number of colors needed to colour a given graph.
- Finding the shortest path between two vertices in a graph.

Decision Problem

- A problem is called a decision problem if its output is a simple "yes" or "no" (or you may need this of this as true/false, 0/1, accept/reject.)
- We will phrase many optimization problems as decision problems.

Example

- Whether a given graph can be colored by only 4-colors.

- Finding Hamiltonian cycle in a graph is not a decision problem, whereas checking a graph is Hamiltonian or not is a decision problem.

CLASSES OF PROBLEMS

1. P-CLASS
2. NP-CLASS
3. NP-HARD
4. NP-COMPLETE

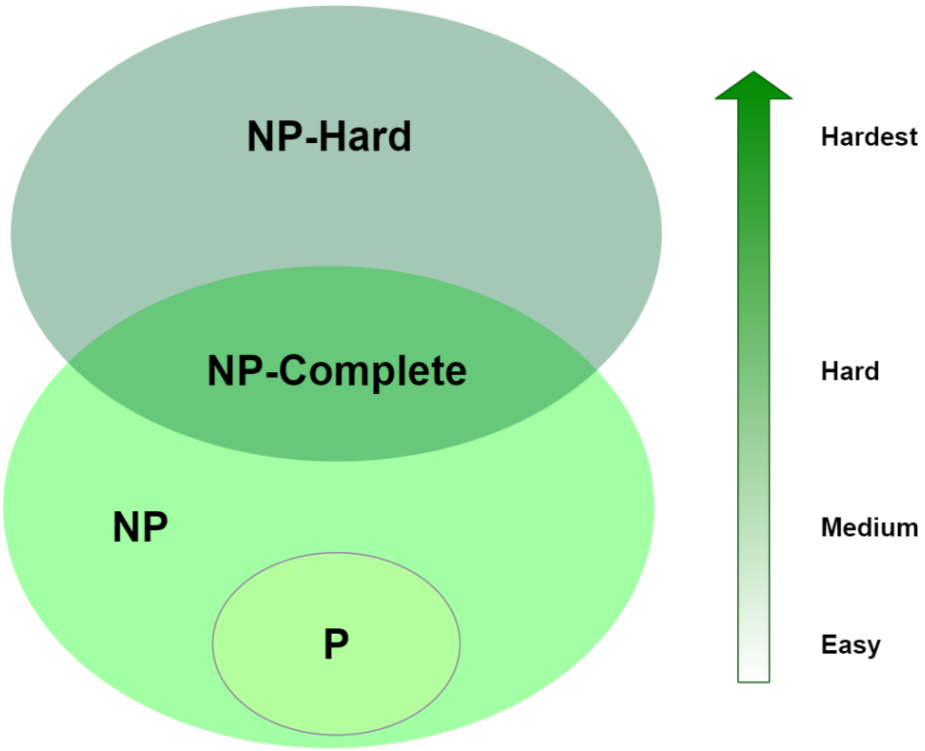
P-Class

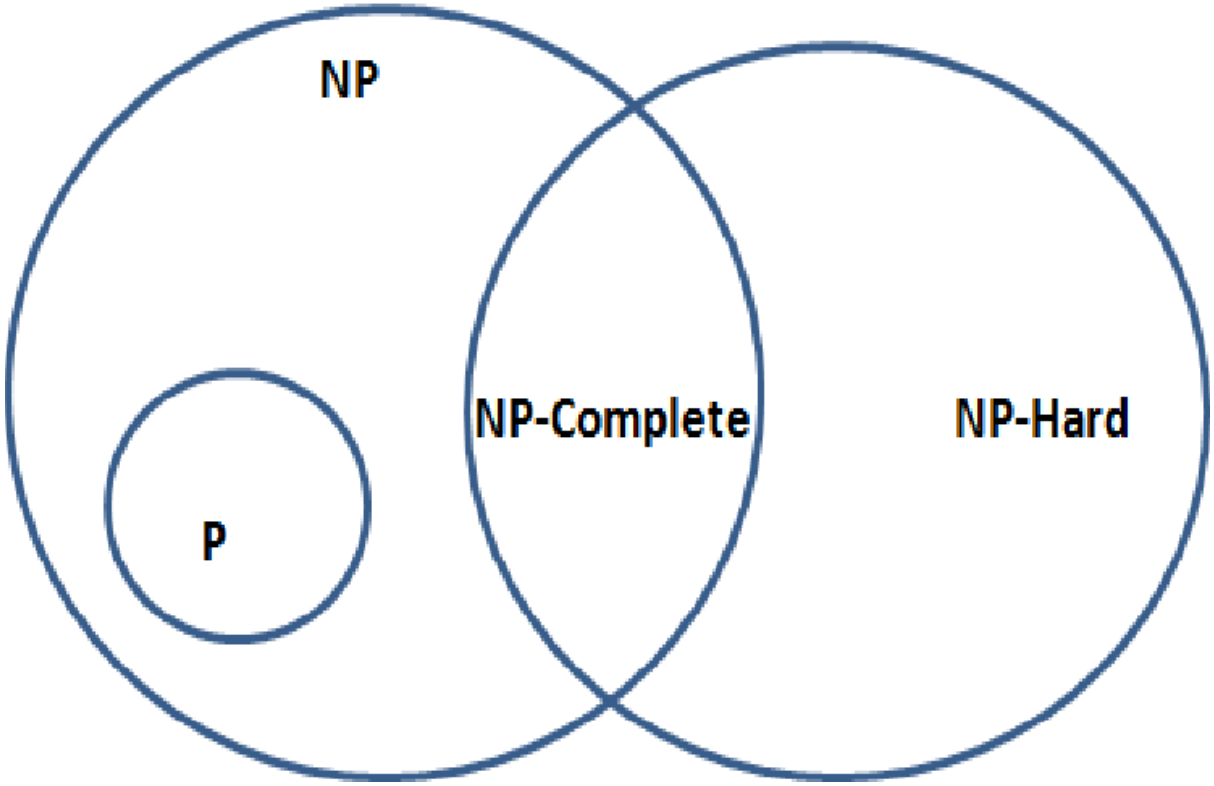
- The set of decision-based problems come into the division of P Problems who can be solved or produced an output within polynomial time.
- These are problems that would be considered 'easy' to solve, and thus do not generally have immense run times.
- The class P consists of those problems that are solvable in polynomial time, i.e. these problems can be solved in time $O(n^k)$ in worst-case, where k is constant.
- These problems are called tractable, while others are called intractable or super polynomial.
- Formally, an algorithm is polynomial time algorithm, if there

exists a polynomial p(n) such that the algorithm can solve any instance of size n in a time O(p(n)).

NP-Class

- The set of all decision-based problems came into the division of NP Problems who can't be solved or produced an output within polynomial time but verified in the **polynomial time**.
- The class NP consists of those problems that are verifiable in polynomial time.
- NP is the class of decision problems for which it is easy to check the correctness of a claimed answer, with the aid of a little extra information.
- NP class problems don’t have a polynomial run-time to *solve*, but have a polynomial run-time to *verify* solutions (difficult to solve, easy to check a given answer).
- NP class contains P class as a subset.
- NP problems being hard to solve.
- The term "NP" does not mean "not polynomial." Originally, the term meant "non-deterministic polynomial”


NP-Hard
NP-Complete
NP
P
Hardest
Hard
Medium
Easy


NP
P
NP-Complete
NP-Hard